

S $\Delta\phi$ -07 — Interpretation-Minimal Operation (H_min): Minimal Axioms and Detection Protocol (v1.0)

Sofience- $\Delta\phi$ Formalism Series

Abstract

This document defines a minimal, operational notion of interpretation (H_min) as an action-level mechanism: marking a path difference and re-entering that mark into subsequent transitions such that the transition distribution changes causally and persistently. The formulation provides three axioms and an empirical detection protocol designed to avoid two common false positives: (i) confusing state compression with interpretation, and (ii) confusing one-off patching with persistent re-entry.

1. Core Definition

Let $\phi(t)$ be the system state at time t , G_t the transition dynamics at time t , and $\Delta\text{path}(t)$ a difference between two possible trajectories or execution paths under comparable starting conditions. Define a marker $m(t)$ as the marking of that path difference: $m(t) = \Pi(\Delta\text{path}(t))$, where Π is a marking (compression) operator. Interpretation-minimal operation is the re-entry of $m(t)$ into the next transition: $\phi(t+1) \sim G_t(\phi(t), m(t))$.

2. Minimal Axioms (with Required Reinforcements)

I1 — Causal Re-entry: for fixed $\phi(t)$, changing the marker changes the next-transition distribution: $P(\phi(t+1)|\phi(t), m1) \neq P(\phi(t+1)|\phi(t), m2)$. A practical criterion is $D_{\text{KL}}(P(.|\phi, m1) || P(.|\phi, m2)) > \delta$.

I2 — Path-Specificity (anti false-positive #1): under the same starting state, different paths must induce different markers, and those marker differences must drive different next-transition distributions. This prevents treating simple state summarization as interpretation.

I3 — Multi-step Persistence (anti false-positive #2): the influence of $m(t)$ must persist for at least two steps, e.g., $\exists k \geq 2$ such that $I(m(t); \phi(t+k) | \phi(t)) > 0$, where $I(\cdot; \cdot | \cdot)$ is conditional mutual information. Practically, marker drop/swap perturbations should produce measurable differences not only at 1-step but also at 2-step outcomes.

3. Minimal Detection Protocol (Operational Test)

- 1 Prepare repeated runs from the same (or equivalent-class) starting state $\phi(t)$.
- 2 Induce path differences $\Delta\text{path}(t)$ by varying exploration, noise, prompts, or action choices while keeping the starting condition comparable.
- 3 Identify a candidate marker $m(t)$ (internal if accessible; otherwise an externalized marker representation).
- 4 Perform marker interventions: (a) keep $m(t)$ intact, (b) drop $m(t)$ (set to null/zero), (c) swap $m(t)$ with a marker from another run, or (d) permute markers across runs.
- 5 Measure next-transition distribution shifts (I1): compare outcomes and compute a divergence metric (e.g., KL) between $P(.|\phi, m)$ conditions.
- 6 Verify path-specificity (I2): under fixed $\phi(t)$, confirm that different induced paths yield different markers and that those markers predict different next transitions.

- 7 Verify persistence (I3): confirm that marker interventions cause measurable differences at 2-step (and beyond) outcomes, not only at 1-step.

4. Series Relations (Recommended)

This document is designed to be used alongside $S\Delta\phi$ -05 (Agency) and $S\Delta\phi$ -06 (Responsibility), as $H_{\min}$ provides a minimal operational interpretation layer for marking and re-entry that can serve as a building block for self-modifying transition-law updates and externality-sensitive constraints.

Recommended Citation

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